Deep Neural Network
Module # 2: Artificial Neural Network

BITS Pilani

Seetha Parameswaran
WILPD - CSIS

innovate achieve lead

The instructor is gratefully acknowledging
the authors who made their course
materials freely available online.
This deck is prepared by Seetha Parameswaran.

Work Integrated Learning Programmes

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

QUESTIONS THAT WILL BE ANSWERED

1. What is an artificial neuron?
2. How is it related to biological neuron?
3. How are artificial neurons interconnected?
4. How do artificial neuron learn?

Work Integrated Learning Programmes

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

How do Humans Learn?

It begins with brain.

- Humans learn, solve problems, recognize patterns, create, think deeply about something, meditate and many many more.....
- Humans learn through association. [Refer to Associationism for more details.]

Work Integrated Learning Programmes

Observation: The Brain

- The brain is a mass of interconnected neurons.
- Number of neurons is approximately 10^{10}.
- Connections per neuron is approximately $10^{(4\ to\ 5)}$.
- Neuron switching time is approximately 0.001 second.
- Scene recognition time is 1 second.
- 100 inference steps doesn't seem like enough.
- Lot of parallel computation.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Biological Neuron Structure

Key Components:

- Dendrites: Receive signals from other neurons
- Cell Body (Soma): Processes incoming signals
- Axon: Transmits output signal
- Synapses: Connection points between neurons

Work Integrated Learning Programmes

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

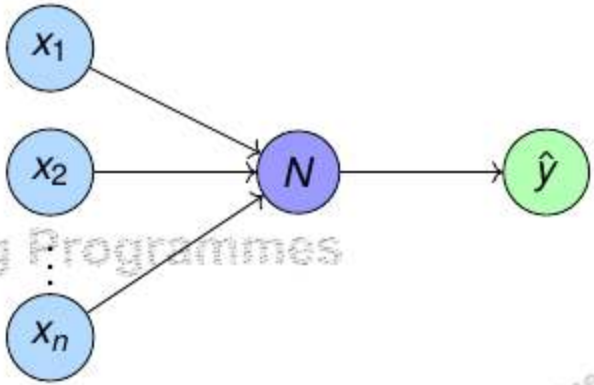
Work Integrated Learning Programmes

What is an Artificial Neuron?

Definition

Artificial Neuron is a processing element inspired by how the brain works.

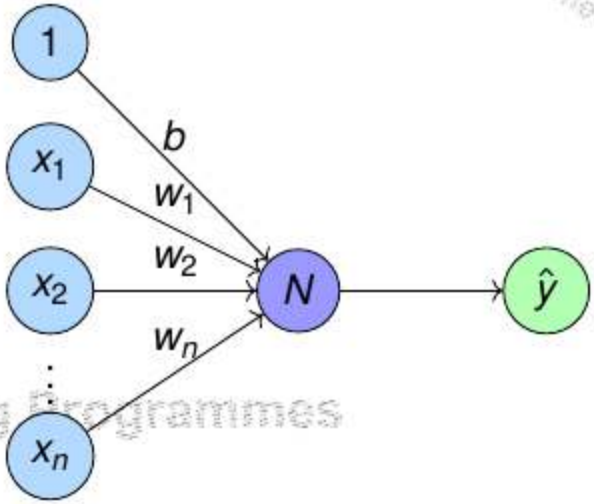
- Similar to biological neuron, each artificial neuron will be do some computation. Each neuron is interconnected to other neurons.
- Similar to brain, the interconnections between neurons store the knowledge it learns. The knowledge is stored as parameters.



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Artificial Neuron N

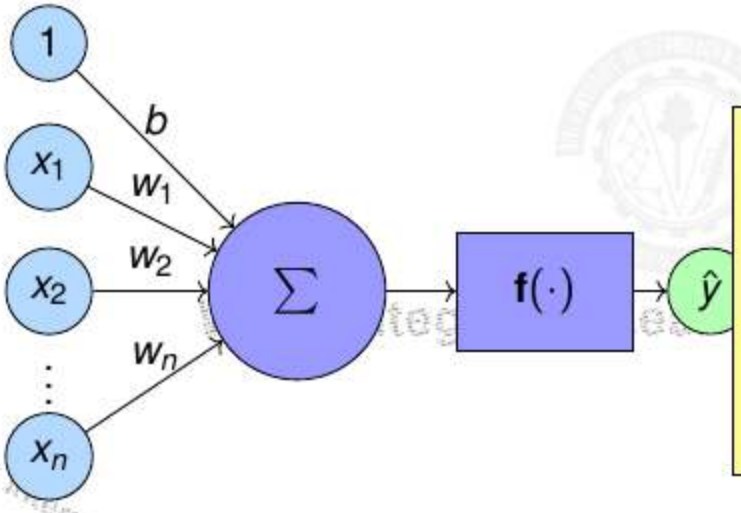
- Processing neuron N performs weighted sum.
- For each feature x_i, weight w_i shows the importance to the feature .
- N generates one numerical output $\hat{y}$



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Artificial Neuron – Mathematical Computation

- Sum of weights w_i times corresponding features x_i
- Apply activation function $f()$ on the sum.



$$z = \sum_{i=1}^{n} w_i x_i + b \qquad (1)$$

$$\hat{y} = \mathbf{f}\left(\sum_{i=1}^{n} w_i x_i + b\right) \qquad (2)$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Comparison: Biological vs Artificial Neurons

Biological Neuron	Artificial Neuron
Complex biochemical processes	Simple mathematical model
Analog signal processing	Digital computation
Adaptive synaptic strengths	Adjustable weights
Parallel processing	Parallel computation possible
Learning through experience	Learning through algorithms
Fault tolerant	Deterministic behavior

Key Insight

Artificial neurons are simplified mathematical abstractions that capture the essential computational properties of biological neurons.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

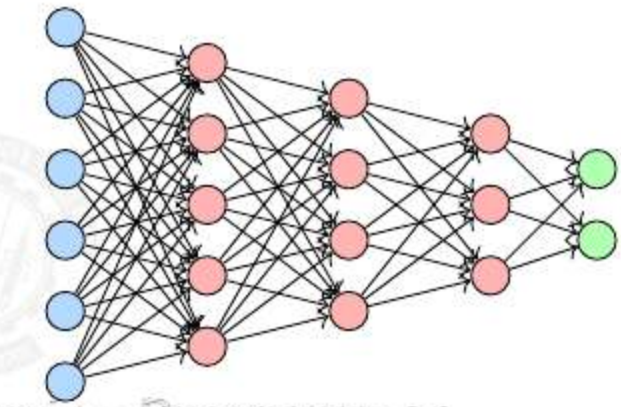
Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

WHAT ARE ARTIFICIAL NEURAL NETWORKS?

- Neural Network is a computational model inspired by human brain.
- Interconnected neurons process information.
- Interconnection can be forward, recursive or self loop.
- Learn patterns from data through training.
- Capable of approximating complex functions



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Properties of Artificial Neural Networks

- Many neurons.
- Many weighted interconnections among neurons.
- Highly parallel, distributed process.
- Tuning parameters or weights automatically.

Work Integrated Learning Programmes

When to consider Artificial Neural Networks?

- Input is high-dimensional discrete or real-valued (e.g. raw sensor input).
- Possibly noisy data. Data has lots of errors.
- Output is discrete or real valued or a vector of values
- Form of target function is unknown.
- Human readability, in other words, explainability, of result is unimportant.
- Examples:
  - Speech recognition
  - Image classification
  - Financial prediction

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

CONNECTIONIST MACHINES

- Network of processing elements, called artificial neurons.
- The neurons are interconnected to form a network.
- Interconnection can be forward, recursive or self loop.
- All world knowledge is stored in the connections between these elements.
- Neural networks are Connectionist machines.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

CONNECTIONISM PRINCIPLES

Core Ideas:

- Intelligence emerges from simple processing neurons
- Knowledge is stored in connection weights
- Learning modifies connection strengths
- Parallel distributed processing

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Table of Contents

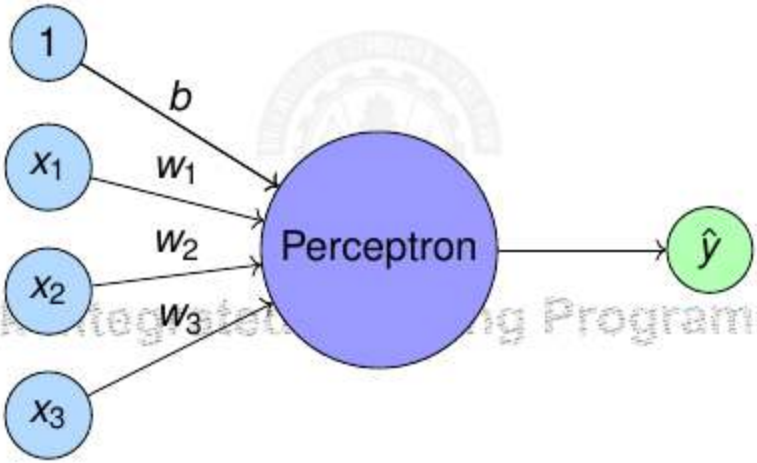
1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

THE PERCEPTRON

Perceptron = Simplest form of artificial neuron.

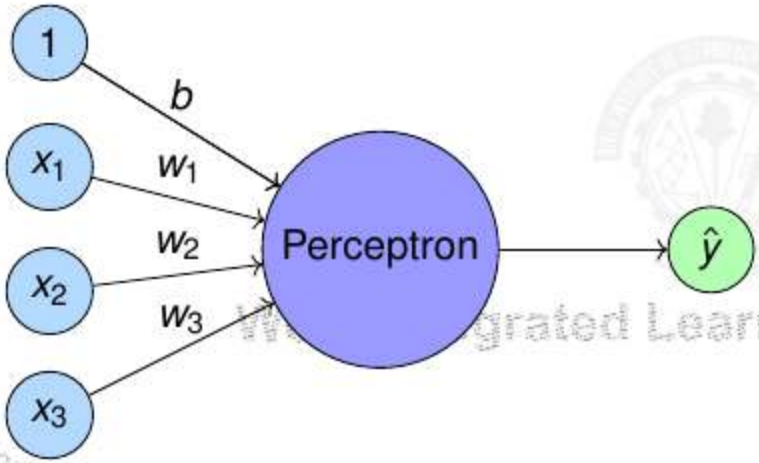
Perceptron was introduced by Rosenblatt in 1958.



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

THE PERCEPTRON

Perceptron = Simplest form of artificial neuron.



$$\hat{y} = \begin{cases} 1 & \text{if } \sum_{i=1}^{n} w_i x_i + b \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

Representing Logic Gates using Perceptron

- Power of Perceptron
  - Any linear decision can be represented by a Boolean equation.
  - The quest is to find out how to represent the above using Perceptron.
  - For this we test whether each of the logic gates can be represented by a Perceptron.
- For the perceptron algorithm:
  - The classical perceptron (Rosenblatt's original) typically uses $\{-1, +1\}$ as outputs.
  - However, modern implementations and logic gates often use $\{0, 1\}$.
  - Both are valid! The choice affects the bias/threshold calculation.

Note: The numerical examples that follow uses binary values $\{0,1\}$. Classical perceptron uses $\{-1,+1\}$, but both formulations are equivalent.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

AND Logic Gate

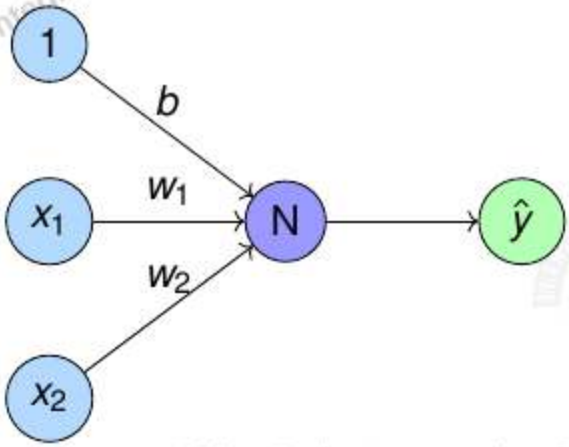
Question:

- How to represent AND gate using a Perceptron?
- What are the parameters for the AND Perceptron?
- Data is given below.

A	B	AB
0	0	0
0	1	0
1	0	0
1	1	1

Work Integrated Learning Programmes

PERCEPTRON FOR AND GATE



Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	0	$w_0 + 0 + w_2 < 0$
1	0	0	$w_0 + w_1 + 0 < 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

$$w_0 = -1$$

$$w_1 = 0.75$$

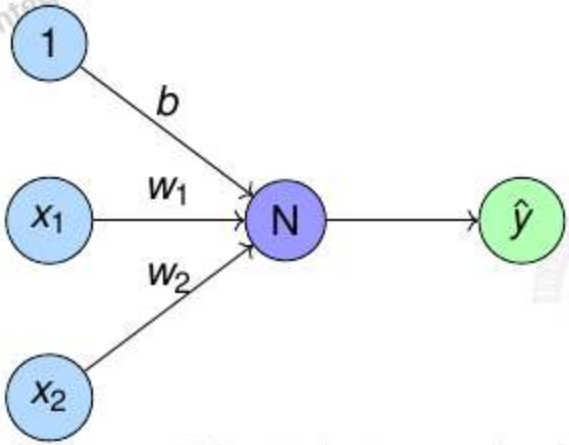
$$w_2 = 0.75$$

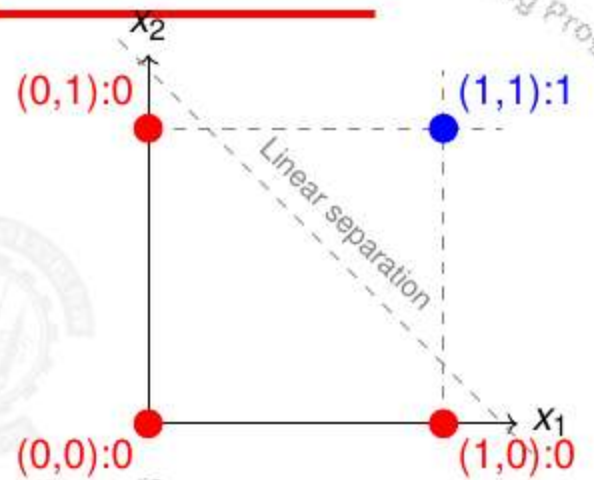
Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

Work Integrated Learning Programmes

PERCEPTRON FOR AND GATE





Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is

$$w_0 = -1$$
$$w_1 = 0.75$$
$$w_2 = 0.75$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

OR Logic Gate

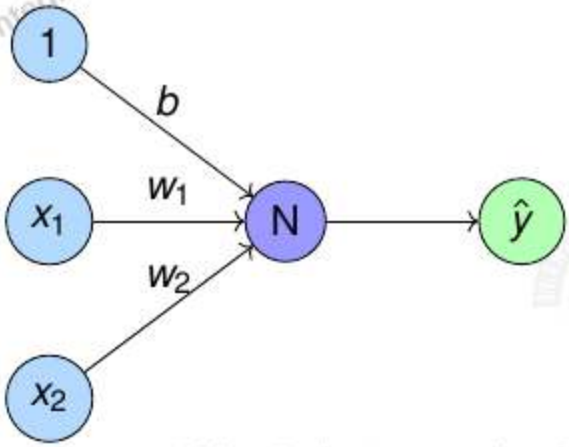
Question:

- How to represent OR gate using a Perceptron?
- What are the parameters for the OR Perceptron?
- Data is given below.

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

Work Integrated Learning Programmes

PERCEPTRON FOR OR GATE



Goal: Find parameters w_0, w_1, w_2

x_1	x_2	t	h
0	0	0	$w_0 + 0 + 0 < 0$
0	1	1	$w_0 + 0 + w_2 \geq 0$
1	0	1	$w_0 + w_1 + 0 \geq 0$
1	1	1	$w_0 + w_1 + w_2 \geq 0$

One solution is

$$w_0 = -1$$

$$w_1 = 2$$

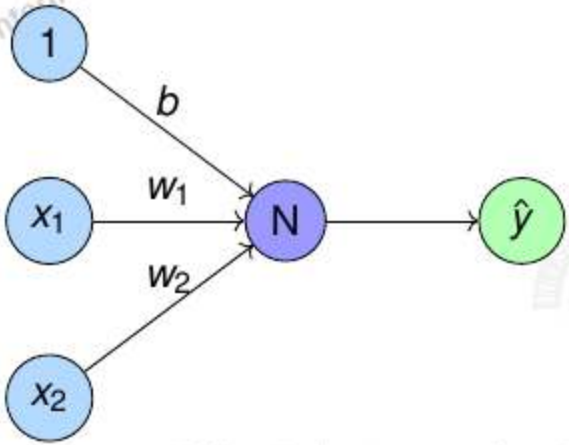
$$w_2 = 2$$

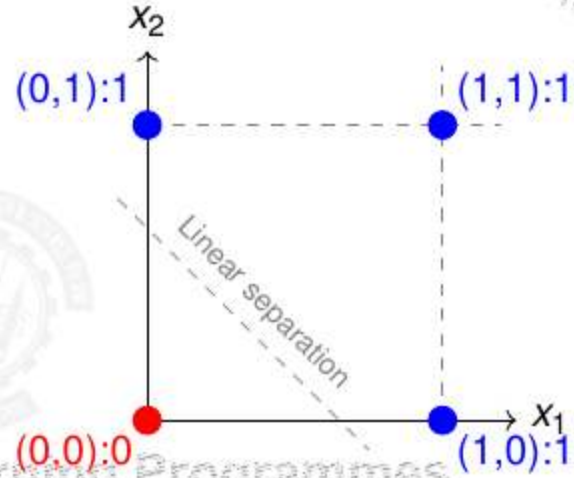
Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

Work Integrated Learning Programmes

PERCEPTRON FOR OR GATE





Perceptron equation is

$$\hat{y} = w_0 x_0 + w_1 x_1 + w_2 x_2$$

One solution is $w_0 = -1$

$w_1 = 2$

$w_2 = 2$

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron Learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron Learning Algorithm

Goal: Find weights that correctly classify training data (input, output) pair (x, t).

Algorithm:

1. Initialize learning rate $\eta = 0.1$
2. Initialize weights randomly: $w_i = \text{random}$
3. For each training example (x, t):
   - Calculate output: $\hat{y} = sign(\sum_i w_i x_i + b)$
   - If $\hat{y} \neq t$, update weights:

$$w_i^{new} = w_i^{old} + \eta(t - \hat{y})x_i$$

$$b^{new} = b^{old} + \eta(t - \hat{y})$$

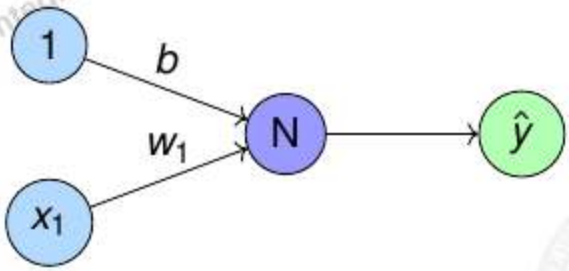
4. Repeat until convergence or max iterations

Convergence Theorem:

If linearly separable data, Perceptron learning algorithm will converge in finite steps.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

PERCEPTRON LEARNING FOR NOT GATE

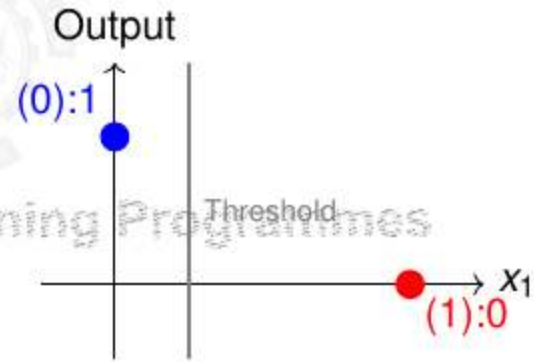


Example	x_1	t
Eg:1	0	1
Eg:2	1	0

Perceptron equation

$$h = w_0 x_0 + w_1 x_1$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases}$$



BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron Learning for NOT Gate I

1. Initialization
   - Assume $w_0 = w_1 = 0$.
   - Let the learning rate$= \eta = 1$.
2. Epoch 1
   - Example 1: $(x_1 = 0, t = 1)$

$$h = w_1 x_1 + w_0 = 0 \cdot 0 + 0 = 0$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases} = 1$$

Is $\hat{y}$ equal to t? YES. No weight update needed.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron Learning for NOT Gate II

- Example 2: $(x_1 = 1, t = 0)$

$$h = w_1 x_1 + w_0 = 0 \cdot 1 + 0 = 0$$
$$\hat{y} = 1$$

Is $\hat{y}$ equal to t? $(\hat{y} = 1, t = 0)$ NO. Update weights:

$$w_i^{(new)} = w_i^{(old)} + \eta(t - \hat{y})x_i$$
$$w_1^{(new)} = 0 + 1(0 - 1) \cdot 1 = -1$$
$$w_0^{(new)} = 0 + 1(0 - 1) = -1$$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron Learning for NOT Gate III

3. Epoch 2 We will use the updated weights $w_1 = -1$, $w_0 = -1$.
   - Example 1: $(x_1 = 0, t = 1)$

$$h = w_1 x_1 + w_0 = (-1) \cdot 0 + (-1) = -1$$
$$\hat{y} = 0 \quad (\text{since } h < 0)$$

Is $\hat{y}$ equal to t ($\hat{y} = 0, t = 1$)? NO. Update weights:

$$w_1^{(new)} = (-1) + 1(1 - 0) \cdot 0 = -1$$
$$w_0^{(new)} = (-1) + 1(1 - 0) = 0$$

   - Example 2: $(x_1 = 1, t = 0)$

$$h = (-1) \cdot 1 + 0 = -1$$
$$\hat{y} = 0$$

Is $\hat{y}$ equal to t? YES. No update needed. Current weights are $w_1 = -1$, $w_0 = 0$

Perceptron Learning for NOT Gate IV

4. Epoch 3
   - Example 1: $(x_1 = 0, t = 1)$

$$h = (-1) \cdot 0 + 0 = 0$$
$$\hat{y} = 1 \quad (\text{since } h \geq 0)$$

   Is $\hat{y}$ equal to t? YES. No update needed.
   - Example 2: $(x_1 = 1, t = 0)$

$$h = (-1) \cdot 1 + 0 = -1$$
$$\hat{y} = 0$$

   Is $\hat{y}$ equal to t? YES. No update needed.

5. Algorithm Converges!

Final weights: $w_1 = -1, w_0 = 0$

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Table of Contents

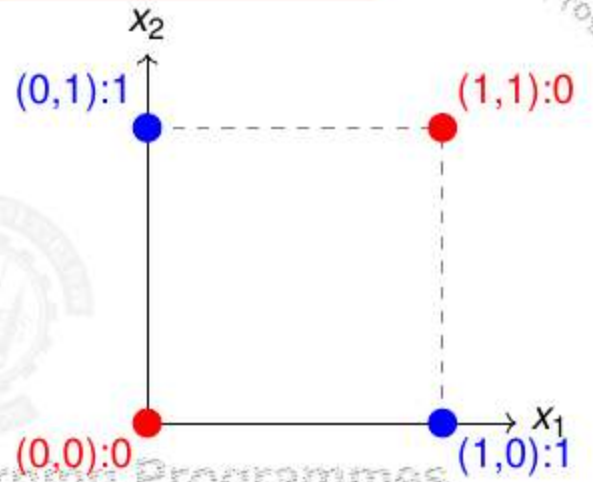
1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

THE XOR PROBLEM

Truth Table for XOR:

x_1	x_2	x_1 XOR x_2
0	0	0
0	1	1
1	0	1
1	1	0



Problem

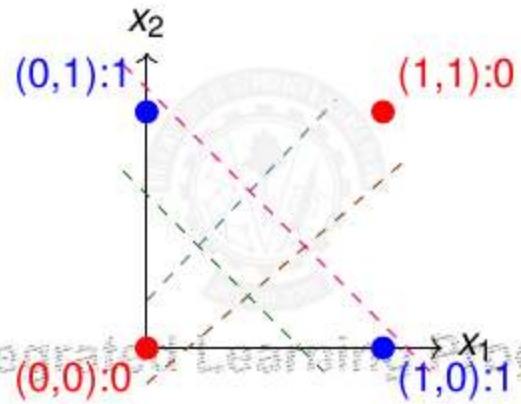
XOR is not linearly separable! Single perceptron cannot solve XOR.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Why Single Perceptron Fails on XOR

Linear Decision Boundary Limitation:

Any line $w_1 x_1 + w_2 x_2 + b = 0$ cannot separate XOR classes.



No single line works!

Solution: Need multiple layers (Multilayer Perceptron)

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

LINEARLY SEPARABLE DATA

- Two sets of data points are linearly separable in a n-dimensional space if they can be separated by an $(n-1)$ dimensional hyperplane.
- in 2D-space, straight line can separate all the data examples belonging to class +1 from all the examples belonging to the class -1. Then it is 2D linearly separable data.
- An infinite number of straight lines can be drawn to separate the class +1 from the class -1.

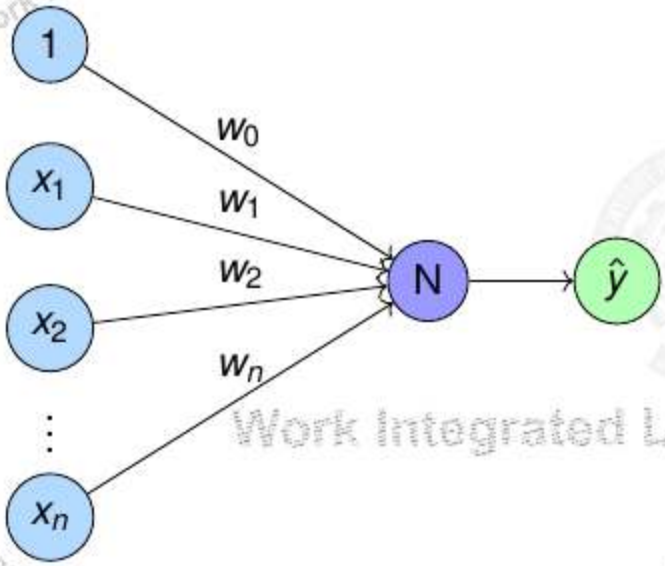
BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron for Linearly Separable Data

- Data
  - Truth tables or set of examples
- Model
  - Perceptron
  - Inputs x are the features or columns
  - Target output t is the required label
- Objective Function
  - Deviation of desired output or target t and the computed output $\hat{y}$
- Learning Algorithm
  - Perceptron Learning algorithm
- Challenge
  - How to learn these $n+1$ parameters $w_0, w_1 . w_2 \ldots w_n$?
- Solution
  - Use Perceptron learning algorithm.

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Perceptron for Linearly Separable Data



$$\tilde{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$h = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

$$= w^\top \tilde{x}$$

$$\hat{y} = \begin{cases} 1 & \text{if } h \geq 0 \\ 0 & \text{if } h < 0 \end{cases}$$

Table of Contents

1. Biological Neuron
2. Artificial Neuron
3. Artificial Neural Network (ANN)
4. Connectionism Model
5. Perceptron
6. Perceptron & Logic Gates
7. How does Perceptron learn?
8. XOR Problem
9. Linearly Separable Data
10. Summary

Work Integrated Learning Programmes

KEY CONCEPTS

- Neural Network: Computational model inspired by biological neurons that processes information through interconnected units
- Perceptron:
  - Single-layer linear classifier: $\hat{y} = \text{sign}(w^\top x)$
  - Learning rule: $w_i^{(new)} = w_i^{(old)} + \eta(t - \hat{y})x_i$
  - Converges for linearly separable data
- Perceptron Limitations:
  - Cannot solve XOR and other non-linearly separable problems
  - Solution: Multi-layer networks with non-linear activation functions
- Foundation: These concepts form the basis for deep learning architectures

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Thank you :)
All the best :)

Work Integrated Learning Programmes